

Determinantal point processes

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CHAPTER 1

Strong Szegő Limit Theorem and Schur measure

1. Statement of the theorem

Throughout, let

$$f(z) = \sum_{k \in \mathbb{Z}} f_k z^k$$

be a sufficiently smooth function on the unit circle. We split f into its nonnegative and negative Fourier parts,

$$f_+(z) := \sum_{k \geq 0} f_k z^k, \quad f_-(z) := \sum_{k < 0} f_k z^k,$$

so that $f = f_+ + f_-$. We write $\tilde{f}(z) := f(z^{-1})$.

THEOREM 1 (Borodin–Okounkov identity). *With f as above,*

$$(1) \quad \det T_n(e^f) = \exp\left(n f_0 + \sum_{k=1}^{\infty} k f_k f_{-k}\right) \det\left(I - Q_n H(e^{f_- - f_+}) H(e^{\tilde{f}_+ - \tilde{f}_-}) Q_n\right).$$

COROLLARY 2 (Strong Szegő limit theorem). *With f as above,*

$$\det T_n(e^f) = \exp\left(n f_0 + \sum_{k=1}^{\infty} k f_k f_{-k}\right) (1 + o(1)), \quad n \rightarrow \infty.$$

We will prove the Borodin–Okounkov identity in several steps. A key input is the following determinant formula.

PROPOSITION 3 (Helton–Howe–Pincus determinant formula). *Let $A, B \in \mathcal{B}(\mathcal{H})$ be bounded operators on a complex Hilbert space such that the commutator*

$$[A, B] := AB - BA$$

is trace class. Then the operators

$$e^{-A} e^{-B} e^{A+B} - I, \quad e^{-A} e^{-B} e^A e^B - I$$

are trace class, and

$$(2) \quad \det(e^{-A} e^{-B} e^{A+B}) = \exp\left(\frac{1}{2} \operatorname{Tr}[A, B]\right),$$

$$(3) \quad \det(e^{-A} e^{-B} e^A e^B) = \exp(\operatorname{Tr}[A, B]).$$

REMARK 4. *At first sight this may look paradoxical: if, say, A were trace class, then cyclicity would imply $\operatorname{Tr}[A, B] = 0$. The point is that neither AB nor BA needs to be trace class, so $\operatorname{Tr}(AB)$ and $\operatorname{Tr}(BA)$ may be undefined, while $\operatorname{Tr}([A, B])$ is well-defined when the commutator is trace class.*

COROLLARY 5. *Let $f(z) = \sum_k f_k z^k$ and $g(z) = \sum_k g_k z^k$ be sufficiently smooth Laurent series. Then*

$$\det(e^{-T(f)} e^{-T(g)} e^{T(f)} e^{T(g)}) = \exp\left(\sum_{k=1}^{\infty} k (g_k f_{-k} - g_{-k} f_k)\right).$$

In particular,

$$\det(e^{-T(f_+)} e^{T(f_-)} e^{T(f_+)} e^{-T(f_-)}) = \exp\left(\sum_{k=1}^{\infty} k f_k f_{-k}\right).$$

PROOF. Using the Toeplitz–Hankel product identity (for sufficiently nice symbols),

$$T(f) T(g) = T(fg) - H(f) H(\tilde{g}),$$

one finds

$$[T(f), T(g)] = H(g) H(\tilde{f}) - H(f) H(\tilde{g}).$$

Both terms are trace class under our smoothness assumptions, and one checks

$$\mathrm{Tr}(H(f) H(\tilde{g})) = \sum_{k=1}^{\infty} k f_k g_{-k}, \quad \mathrm{Tr}(H(g) H(\tilde{f})) = \sum_{k=1}^{\infty} k g_k f_{-k}.$$

Applying (3) to $A = T(f)$ and $B = T(g)$ yields the first identity. For the second, take $f = f_+$ and $g = -f_-$. $\square$

We will also need a standard linear algebra lemma.

LEMMA 6 (Jacobi identity for principal minors). *Let A be an invertible $n \times n$ matrix, and let $I \subset \{1, \dots, n\}$ with complement I^c . Denote by A_I and A_{I^c} the corresponding principal submatrices of A , and by $(A^{-1})_{I^c}$ the principal submatrix of A^{-1} indexed by I^c . Then*

$$\det A_I = (\det A) \det (A^{-1})_{I^c}.$$

PROOF OF THEOREM 1. Set $a = e^f$ for the Toeplitz symbol. Since $T(f_+)$ is lower triangular and $T(f_-)$ is upper triangular, the exponentials $e^{-T(f_{\pm})}$ remain triangular, and hence

$$\det(P_n e^{-T(f_+)} P_n) = e^{-nf_0}, \quad \det(P_n e^{-T(f_-)} P_n) = 1.$$

Therefore,

$$e^{-nf_0} \det T_n(e^f) = \det(P_n e^{-T(f_+)} P_n) \det(P_n T_n(e^f) P_n) \det(P_n e^{-T(f_-)} P_n),$$

and thus

$$e^{-nf_0} \det T_n(e^f) = \det(P_n e^{-T(f_+)} T(e^f) e^{-T(f_-)} P_n).$$

Next we use the Wiener–Hopf factorization at the level of symbols:

$$e^f = e^{f_-} e^{f_+}, \quad \text{so that} \quad T(e^f) = T(e^{f_-}) T(e^{f_+}).$$

(Here e^{f_-} has only nonpositive modes and e^{f_+} only nonnegative modes.) Moreover, since $T(e^{f_{\pm}}) = e^{T(f_{\pm})}$ for triangular Toeplitz operators,

$$T(e^{f_-}) = e^{T(f_-)}, \quad T(e^{f_+}) = e^{T(f_+)}.$$

Hence

$$e^{-nf_0} \det T_n(e^f) = \det(P_n e^{-T(f_+)} e^{T(f_-)} P_n e^{T(f_+)} P_n e^{-T(f_-)} P_n).$$

By the triangular structure we can drop the middle P'_n s and arrive at

$$e^{-nf_0} \det T_n(e^f) = \det \left(P_n e^{-T(f_+)} e^{T(f_-)} e^{T(f_+)} e^{-T(f_-)} P_n \right).$$

Applying the Jacobi identity (principal minors) yields

$$e^{-nf_0} \det T_n(e^f) = \det \left(Q_n e^{T(f_-)} e^{-T(f_+)} e^{-T(f_-)} e^{T(f_+)} Q_n \right) \det \left(e^{-T(f_+)} e^{T(f_-)} e^{T(f_+)} e^{-T(f_-)} \right).$$

By Corollary 5, the second determinant equals

$$\exp \left(\sum_{k=1}^{\infty} k f_k f_{-k} \right).$$

Multiplying back the factor e^{nf_0} gives

$$\det T_n(e^f) = \exp \left(nf_0 + \sum_{k=1}^{\infty} k f_k f_{-k} \right) \det \left(Q_n e^{T(f_-)} e^{-T(f_+)} e^{-T(f_-)} e^{T(f_+)} Q_n \right).$$

Finally, using again the Toeplitz–Hankel product identity, one checks that

$$e^{T(f_-)} e^{-T(f_+)} e^{-T(f_-)} e^{T(f_+)} = I - H(e^{f_- - f_+}) H(e^{\tilde{f}_+ - \tilde{f}_-}),$$

which gives (1). $\square$

2. Partition function for the biased Aztec diamond

We start from the following observation.

THEOREM 7. *Let $a \in (0, 1)$ be the bias parameter in the Aztec diamond model, where horizontal dominos have weight 1 and vertical dominos have weight a . Then the partition function Z_N of the Aztec diamond of size N is given by*

$$Z_N = (1 + a^2)^{N^2}.$$

To prove this result we first establish the following lemma.

LEMMA 8. *Define the Laurent polynomial*

$$\phi(z) = (1 + az)^N (1 - a/z)^N.$$

Then, for all $n \geq N$,

$$\det T_n(\phi) = Z_N Z_{N-1}.$$

PROOF. By the Lindström–Gessel–Viennot theorem, the partition function of the non-intersecting path ensemble consisting of n paths that make $2N$ steps, with transition matrices alternating between $L(1 + az)$ and $L(1/(1 - a/z))$, and starting and ending at $(0, -j)$ for $j = 1, \dots, n$, is given by $\det T_n(\phi)$.

The corresponding tiling model decomposes into three independent regions: an Aztec diamond of size N , a frozen corridor, and an Aztec diamond of size $N - 1$. Since no domino can intersect more than one region, the partition function factorizes into the product of the partition functions of these three regions. The frozen corridor contributes a factor 1, which yields the stated identity. $\square$

PROOF OF THEOREM 7. We apply the Strong Szegő limit theorem. Write $\phi = e^f$, where

$$f(z) = N \log(1 + az) + N \log(1 - a/z).$$

Then $f_0 = 0$, and for $k \geq 1$,

$$f_k = \frac{N(-1)^{k-1}a^k}{k}, \quad f_{-k} = -\frac{Na^k}{k}.$$

Hence,

$$\sum_{k=1}^{\infty} k f_k f_{-k} = N^2 \sum_{k=1}^{\infty} (-1)^k \frac{a^{2k}}{k} = N^2 \log(1 + a^2).$$

The Strong Szegő limit theorem gives

$$\lim_{n \rightarrow \infty} \det T_n(\phi) = (1 + a^2)^{N^2}.$$

Now, for N odd, write

$$Z_N = Z_1 \frac{Z_3 Z_2}{Z_1 Z_2} \frac{Z_5 Z_4}{Z_3 Z_5} \cdots = Z_1 \prod_{j=1}^{N/2-1} \frac{Z_{2j+1} Z_{2j}}{Z_{2j} Z_{2j-1}}$$

and use that $Z_1 = (1 + a^2)$. Thus,

$$Z_N = (1 + a^2) \prod_{j=1}^{N/2-1} \frac{(1 + a^2)^{(2j+1)^2}}{(1 + a^2)^{(2j)^2}} = (1 + a^2)^{1 + \sum_{j=1}^{N/2-1} (2j+1)^2 - (2j)^2} = (1 + a^2)^{N(N-1)/2}.$$

If N is even one writes a similar telescoping argument and use that $Z_0 = 1$. $\square$

3. Connections to the Schur measure

3.1. Schur functions and determinantal identities.

3.1.1. *Partitions.* A *partition* is a finite non-increasing sequence of nonnegative integers

$$\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0),$$

where $r = \ell(\lambda)$ is the *length* of λ and $|\lambda| = \sum_i \lambda_i$ its *size*. We allow trailing zeros and identify partitions that differ only by zeros. The *conjugate* partition λ' is defined by

$$\lambda'_j := \#\{i : \lambda_i \geq j\}.$$

3.1.2. *Symmetric polynomials in N variables.* Let $x = (x_1, \dots, x_N)$. We write $\Lambda_N = \mathbb{C}[x_1, \dots, x_N]^{S_N}$ for the ring of symmetric polynomials in N variables.

3.1.3. *Complete and elementary symmetric polynomials.* The *complete homogeneous symmetric polynomials* $h_k(x)$ are defined by

$$h_k(x_1, \dots, x_N) = \sum_{1 \leq i_1 \leq \cdots \leq i_k \leq N} x_{i_1} \cdots x_{i_k}, \quad k \geq 1,$$

with the conventions $h_0 = 1$ and $h_k = 0$ for $k < 0$. Equivalently, their generating function is

$$(4) \quad H(x; t) := \sum_{k \geq 0} h_k(x) t^k = \prod_{i=1}^N \frac{1}{1 - x_i t}.$$

The *elementary symmetric polynomials* $e_k(x)$ are defined by

$$e_k(x_1, \dots, x_N) = \sum_{1 \leq i_1 < \cdots < i_k \leq N} x_{i_1} \cdots x_{i_k}, \quad k \geq 1,$$

with $e_0 = 1$ and $e_k = 0$ for $k < 0$, and generating function

$$(5) \quad E(x; t) := \sum_{k \geq 0} e_k(x) t^k = \prod_{i=1}^N (1 + x_i t).$$

Note that $H(x; t) E(x; -t) = 1$.

3.1.4. *Schur functions in N variables.* Let λ be a partition with $\ell(\lambda) \leq N$. The *Schur polynomial* (or Schur function in N variables) is defined by the alternant ratio

$$(6) \quad s_\lambda(x_1, \dots, x_N) := \frac{\det(x_i^{\lambda_j + N - j})_{1 \leq i, j \leq N}}{\det(x_i^{N - j})_{1 \leq i, j \leq N}}.$$

The denominator is the Vandermonde determinant

$$\det(x_i^{N - j})_{i, j=1}^N = \prod_{1 \leq i < j \leq N} (x_i - x_j),$$

hence s_λ is a symmetric polynomial. If $\ell(\lambda) > N$ we set $s_\lambda(x_1, \dots, x_N) = 0$.

It will be useful to have the following notation.

$$\det(x_i^{\lambda_j + N - j})_{1 \leq i, j \leq N} = a_\lambda(x).$$

3.1.5. *Generators and an involution.* The ring Λ of symmetric functions is a commutative ring with unit. Two particularly important generating families are given by the complete and elementary symmetric functions.

The complete symmetric functions $\{h_k\}_{k \geq 1}$ are algebraically independent and generate Λ as a polynomial ring:

$$\Lambda \cong \mathbb{C}[h_1, h_2, h_3, \dots].$$

Equivalently, every symmetric function can be written uniquely as a polynomial with in the variables h_k . An entirely analogous statement holds for the elementary symmetric functions $\{e_k\}_{k \geq 1}$,

$$\Lambda \cong \mathbb{C}[e_1, e_2, e_3, \dots].$$

These two generating sets are related by a canonical involutive ring automorphism

$$\omega : \Lambda \rightarrow \Lambda, \quad \omega(h_k) = e_k,$$

which extends uniquely to all of Λ . The involution ω exchanges the two natural systems of generators and reflects a fundamental duality in symmetric function theory; in particular, it sends Schur functions to their conjugates, $\omega(s_\lambda) = s_{\lambda'}$.

The Schur functions form a basis for Λ_N .

3.1.6. *The Cauchy identities.*

THEOREM 9. *We have the following identities*

- (Cauchy Identity) $\sum_{\lambda} s_{\lambda}(x) s_{\lambda}(y) = \prod_{i,j=1}^N \frac{1}{1 - x_i y_j}$.
- (Dual Cauchy Identity) $\sum_{\lambda} s'_{\lambda}(x) s_{\lambda}(y) = \prod_{i,j=1}^N (1 + x_i y_j)$.

PROOF. The first identity follows from the following relation:

$$\det \left(\frac{1}{1 - x_i y_j} \right)_{i,j=1}^N = \frac{\prod_{1 \leq i < j \leq N} (x_i - x_j) \prod_{1 \leq i < j \leq N} (y_i - y_j)}{\prod_{i,j=1}^N (1 - x_i y_j)}.$$

The proof of this relation goes as follows: multiply both sides with $\prod_{i,j=1}^N (1 - x_i y_j)$. Then the left hand-side is a symmetric polynomial in $x_1, \dots, x_N$ and also

a symmetric polynomial in $y_1, \dots, y_N$. It also vanishes if $x_i = x_j$ or $y_i - y_j$ for $i \neq j$. Hence the right-hand side is a factor. It just remains to see that there are not further factors but this can be seen by comparing the degree. We are thus left with proving

$$\det \left(\frac{1}{1 - x_i y_j} \right)_{i,j=1}^N = \sum_{\lambda} a_{\lambda}(x) a_{\lambda}(y).$$

This however follows after a tedious computation expanding the determinant on the left and then expanding each entry as a geometric series $(1 - r)^{-1} = \sum_{k \geq 0} r^k$. We leave the details to the reader.

The second identity follows from the first after invoking the involution ω on $s_{\lambda}(x)$. $\square$

3.1.7. Jacobi–Trudi identities.

THEOREM 10 (Jacobi–Trudi). *Let λ be a partition with $\ell(\lambda) \leq r \leq N$. Then*

$$(7) \quad s_{\lambda}(x_1, \dots, x_N) = \det(h_{\lambda_i - i + j}(x_1, \dots, x_N))_{1 \leq i, j \leq r}.$$

Moreover, letting λ' be the conjugate partition, one has the dual form, for any $\lambda_1 \leq r \leq N$

$$(8) \quad s_{\lambda}(x_1, \dots, x_N) = \det(e_{\lambda'_i - i + j}(x_1, \dots, x_N))_{1 \leq i, j \leq r}.$$

PROOF. We start with the first identity. For $f(y) = \sum_{j \geq 0} f_j y^j$, Cauchy–Binet identity and a quick rewriting gives

$$f(y_1) \cdots f(y_N) = \det(x_i^{N-j})_{i,j=1}^N s_{\lambda}(y_1, \dots, y_N).$$

Apply this with $f(y) = \prod_{i=1}^N 1/(1 - x_i y)$, use the Cauchy identity and the fact that s_{λ} for a basis.

Now we come to the second identity. Consider the standard involution ω of the ring of symmetric functions characterized (in N variables) by $\omega(h_k) = e_k$. On Schur polynomials it satisfies

$$\omega(s_{\lambda}) = s_{\lambda'}.$$

Applying ω to (7) gives

$$s_{\lambda'} = \det(e_{\lambda'_i - i + j})_{1 \leq i, j \leq r'}, \quad r' \geq \ell(\lambda').$$

Replacing λ by λ' and noting that $\ell(\lambda') = \lambda_1$ yields (8). $\square$

3.1.8. Gessel’s identity.

Let $x = (x_1, \dots, x_N)$ and $y = (y_1, \dots, y_N)$. Define

$$E(x; z) := \sum_{k \geq 0} e_k(x) z^k = \prod_{i=1}^N (1 + x_i z), \quad E(y; z) := \sum_{k \geq 0} e_k(y) z^k = \prod_{j=1}^N (1 + y_j z),$$

and set the Laurent symbol

$$(9) \quad \psi(z) := E(x; z) E(y; z^{-1}) = \left(\prod_{i=1}^N (1 + x_i z) \right) \left(\prod_{j=1}^N (1 + y_j z^{-1}) \right) = \sum_{k \in \mathbb{Z}} \psi_k z^k.$$

Let $T_n(\psi) = (\psi_{j-k})_{0 \leq j, k \leq n-1}$ be the $n \times n$ Toeplitz matrix.

THEOREM 11 (Gessel). *With ψ as in (9),*

$$(10) \quad \det T_n(\psi) = \sum_{\lambda: \lambda_1 \leq n} s_{\lambda}(x) s_{\lambda}(y).$$

PROOF. Expand

$$E(x; z) = \sum_{k \geq 0} e_k(x) z^k, \quad E(y; z^{-1}) = \sum_{\ell \geq 0} e_\ell(y) z^{-\ell},$$

so for $m \in \mathbb{Z}$,

$$\psi_m = \sum_{\ell \geq 0} e_{m+\ell}(x) e_\ell(y), \quad (e_k = 0 \text{ for } k < 0).$$

Define the $n \times \infty$ matrices (rows $0, \dots, n-1$, columns $\ell \geq 0$)

$$(\tilde{A}_n(x))_{j,\ell} := e_{j+\ell}(x), \quad (\tilde{A}_n(y))_{j,\ell} := e_{j+\ell}(y).$$

A direct index check gives the factorization

$$T_n(\psi) = \tilde{A}_n(x) \tilde{A}_n(y)^T.$$

By Cauchy–Binet,

$$\det T_n(\psi) = \sum_{\substack{J \subset \mathbb{Z}_{\geq 0} \\ |J|=n}} \det(\tilde{A}_n(x)_{[0,n-1],J}) \det(\tilde{A}_n(y)_{[0,n-1],J}).$$

Parametrize $J = \{j_0 < \dots < j_{n-1}\}$ by a partition λ with $\lambda_1 \leq n$ via

$$j_{i-1} = \lambda'_i + i - 1, \quad i = 1, \dots, n,$$

so that $\lambda'_i = j_{i-1} - (i-1)$ and $\ell(\lambda') \leq n$, i.e. $\lambda_1 \leq n$. With this choice, the minors are dual Jacobi–Trudi determinants:

$$\det(\tilde{A}_n(x)_{[0,n-1],J}) = \det(e_{\lambda'_i - i + j}(x))_{1 \leq i, j \leq n} = s_\lambda(x),$$

and similarly for y . Substituting yields (10). $\square$

3.2. The Schur measure. Let $x = (x_1, \dots, x_N)$ and $y = (y_1, \dots, y_M)$ be two sets of variables such that

$$Z := \sum_{\lambda} s_\lambda(x) s_\lambda(y) < \infty,$$

for instance when $|x_i|, |y_j| < 1$. The *Schur measure* on the set of partitions is defined by

$$(11) \quad \mathbb{P}(\lambda) = \frac{1}{Z} s_\lambda(x) s_\lambda(y).$$

where

$$(12) \quad Z = \sum_{\lambda} s_\lambda(x) s_\lambda(y)$$

We now return to the non-intersecting path model described earlier. By the Jacobi–Trudi relation we can regard the Schur measure as a measure on non-intersecting paths. Indeed, consider n paths that start at $(0, -j)$ and end at $(M+N, -j)$. They live on a graph that consist of N type III columns glued to M type IV columns. The transition for the type III columns are given by

$$T_{m,m+1} = L(\tilde{a}_{m,m+1})$$

with $a_m(z) = 1/(1 - x_m z)$, for $m = 0, \dots, N-1$ and

$$T_{N+m,N+m+1} = L(\tilde{a}_{m,m+1})$$

with $a_{N+m}(z) = 1/(1 - y_m/z)$, for $m = 0, \dots, M-1$.

Then let p_j the position of the paths after N steps. Then it follows that the Jacobi-Trudi relation gives that

$$(p_j)_j = (\lambda_j - j)_j,$$

in law.

It is thus also clear from our previous analysis that Schur measures (in these variables) form a determinantal point process and that the kernel can be found by computing the Wiener-Hopf factorization of $a(z) = \prod_{m=0}^{N-1} a_{m,m+1}(z)$. This factorization now take

$$a_+(z) = H(x; z), \quad a_-(z) = H(y, -).$$

Then the kernel takes the form

$$K(p, p') = \frac{1}{(2\pi i)^2} \oint_{|z|=r_-} dz \oint_{|w|=r_+} dw \frac{a_-(z)a_+(w)}{a_+(z)a_-(w)} \frac{1}{w(w-z)} \frac{z^p}{w^{p'}},$$

3.3. Partition function: the Cauchy identity via Strong Szegő. We compute the partition function for the Schur measure by invoking the strong Szegő limit theorem.

THEOREM 12. [Cauchy-Identity] *Let $x = (x_1, \dots, x_N)$ and $y = (y_1, \dots, y_M)$ satisfy $|x_i| < 1$ and $|y_j| < 1$. Then*

$$Z = \sum_{\lambda} s_{\lambda}(x) s_{\lambda}(y) = \prod_{i=1}^N \prod_{j=1}^M \frac{1}{1 - x_i y_j}.$$

PROOF. Set

$$H(x; z) = \prod_{i=1}^N \frac{1}{1 - x_i z}, \quad H(y; z) = \prod_{j=1}^M \frac{1}{1 - y_j z}, \quad \phi(z) = H(x; z) H(y; z^{-1}).$$

Let $T_n(\phi)$ be the $n \times n$ Toeplitz matrix with symbol ϕ .

Step 1: Toeplitz determinant expansion. By Cauchy–Binet together with the Jacobi–Trudi identity (equivalently, the Toeplitz-minor representation of s_{λ}), one has for every n ,

$$\det T_n(\phi) = \sum_{\lambda: \ell(\lambda) \leq n} s_{\lambda}(x) s_{\lambda}(y).$$

Since $s_{\lambda}(x) = 0$ for $\ell(\lambda) > N$, the right-hand side stabilizes for $n \geq N$, hence

$$\lim_{n \rightarrow \infty} \det T_n(\phi) = \sum_{\lambda} s_{\lambda}(x) s_{\lambda}(y).$$

Step 2: Strong Szegő. Write $\phi = e^a$ with

$$a(z) = - \sum_{i=1}^N \log(1 - x_i z) - \sum_{j=1}^M \log(1 - y_j z^{-1}).$$

Then $a_0 = 0$ and for $k \geq 1$,

$$a_k = \frac{1}{k} \sum_{i=1}^N x_i^k, \quad a_{-k} = \frac{1}{k} \sum_{j=1}^M y_j^k.$$

By the Strong Szegő limit theorem,

$$\lim_{n \rightarrow \infty} \det T_n(\phi) = \exp\left(\sum_{k \geq 1} k a_k a_{-k}\right).$$

Compute

$$\sum_{k \geq 1} k a_k a_{-k} = \sum_{i=1}^N \sum_{j=1}^M \sum_{k \geq 1} \frac{(x_i y_j)^k}{k} = - \sum_{i=1}^N \sum_{j=1}^M \log(1 - x_i y_j),$$

and therefore

$$\lim_{n \rightarrow \infty} \det T_n(\phi) = \prod_{i=1}^N \prod_{j=1}^M \frac{1}{1 - x_i y_j}.$$

Combining the two steps yields the claim. $\square$

It is important to note that we already had a proof of the Cauchy identity. This means that we can also read the above proof in the reverse direction. Which means that the Cauchy identity is a means to prove the strong Szegő Limit Theorem. As we will see next, the Borodin-Okounkov identity can also be obtained in this way.

3.4. Revisiting the Borodin-Okounkov Formula. The event $\{\lambda_1 \leq n\}$ is equivalent (under our particle convention) to the absence of particles in $\{n, n+1, \dots\}$. Therefore,

$$(13) \quad \frac{1}{Z} \sum_{\lambda: \lambda_1 \leq n} s_\lambda(x) s_\lambda(y) = \det(I - K_n).$$

Now let us rewrite K_n a bit:

$$(14) \quad \begin{aligned} K(p, p') &= \frac{1}{(2\pi i)^2} \oint_{|z|=r_-} dz \oint_{|w|=r_+} dw \frac{a_-(z) a_+(w)}{a_+(z) a_-(w)} \frac{1}{w(w-z)} \frac{z^p}{w^{p'}} \\ &= \sum_{\ell=0}^{\infty} \frac{1}{(2\pi i)^2} \oint_{|z|=r_-} dz \oint_{|w|=r_+} dw \frac{a_-(z) a_+(w)}{a_+(z) a_-(w)} \frac{1}{wz} \frac{z^{p+\ell+1}}{w^{p'+\ell+1}} \\ &= (H(a_+/a_-) H(\tilde{a}_-/\tilde{a}_+))_{p', p}. \end{aligned}$$

Thus

$$\det(I - K_n) = \det(I - K_n^T) = \det(I - Q_n H(a_+/a_-) H(\tilde{a}_-/\tilde{a}_+) Q_n)$$

By the Borodin-Okounkov Theorem this is equal to

$$Z^{-1} \det T_n(a^{-1}).$$

(note that we have used $Z(a) = Z(a^{-1})$.) Together, this implies that

$$\sum_{\lambda: \lambda_1 \leq n} s_\lambda(x) s_\lambda(y) = \det T_n(a^{-1}).$$

But this is precisely Gessel's identity!

REMARK 13 (Historical comment). *The argument above can also be read in reverse. Assuming the Cauchy identity and Gessel's identity for Schur functions, one can retrace the steps to obtain the Borodin-Okounkov formula for Toeplitz determinants. In fact, this was essentially the original route by which the identity*

was discovered: Borodin and Okounkov derived their formula by exploiting determinantal identities arising from Schur measures, rather than from operator-theoretic considerations.

Shortly thereafter, several independent proofs based on operator theory, Wiener–Hopf factorization, and trace class commutator identities were found, clarifying the analytic structure behind the formula. In hindsight, it also became clear that closely related identities had appeared earlier in the work of Geronimo and Case on Toeplitz determinants and scattering theory. For this reason, the identity is sometimes more accurately referred to as the Borodin–Okounkov–Geronimo–Case identity, although the shorter name “Borodin–Okounkov identity” has become standard in the literature.

APPENDIX A

Laurent, Toeplitz, and Hankel operators

0.1. Laurent operators and symbols. Let $\ell^2(\mathbb{Z})$ be the Hilbert space of square-summable sequences $f = (f_k)_{k \in \mathbb{Z}}$. A (bi-infinite) matrix $L = (L_{ij})_{i,j \in \mathbb{Z}}$ is called a *Laurent matrix* if L_{ij} depends only on $i - j$, i.e.

$$L_{ij} = a_{i-j} \quad (i, j \in \mathbb{Z})$$

for some sequence $(a_k)_{k \in \mathbb{Z}}$. The corresponding bounded operator on $\ell^2(\mathbb{Z})$ (when it exists) is called a *Laurent operator* and is denoted $L(a)$, where a is the *symbol*

$$a(z) = \sum_{k \in \mathbb{Z}} a_k z^k, \quad z \in \mathbb{T}.$$

When $(a_k) \in \ell^1(\mathbb{Z})$, the series converges absolutely on $\mathbb{T}$ and $L(a)$ is bounded, acting by convolution:

$$(L(a)f)_i = \sum_{k \in \mathbb{Z}} a_{i-k} f_k.$$

In this case one has the algebra property

$$L(a)L(b) = L(ab),$$

reflecting the fact that Laurent operators diagonalize under the Fourier transform.

A convenient Banach algebra for symbols is the *Wiener algebra*

$$\mathcal{W} := \left\{ a(z) = \sum_{k \in \mathbb{Z}} a_k z^k : \sum_{k \in \mathbb{Z}} |a_k| < \infty \right\}, \quad \|a\|_{\mathcal{W}} = \sum_{k \in \mathbb{Z}} |a_k|.$$

If $a \in \mathcal{W}$ and $a(z) \neq 0$ for all $z \in \mathbb{T}$, then $1/a \in \mathcal{W}$ (Wiener's lemma). Consequently,

$$L(a)^{-1} = L(1/a)$$

on $\ell^2(\mathbb{Z})$. This is one reason why operator-level inverses are often simpler than finite-matrix inverses.

0.2. Toeplitz and Hankel operators. Let $\ell^2(\mathbb{Z}_{\geq 0})$ be the Hardy-type half-space, and let $P_+ : \ell^2(\mathbb{Z}) \rightarrow \ell^2(\mathbb{Z})$ be the orthogonal projection onto indices ≥ 0 . For a Laurent operator $L(a)$, the associated *Toeplitz operator* is the compression

$$T(a) := P_+ L(a) P_+ \Big|_{\ell^2(\mathbb{Z}_{\geq 0})}.$$

In matrix form, $T(a)$ has entries

$$T(a)_{ij} = a_{i-j} \quad (i, j \geq 0),$$

so it is constant along diagonals. The *finite Toeplitz matrix* of size n is the further compression

$$T_n(a) := P_n T(a) P_n,$$

where P_n projects onto $\{0, 1, \dots, n-1\}$.

The *Hankel operator* associated with a is defined by

$$H(a) := P_+ L(a) J P_+ \Big|_{\ell^2(\mathbb{Z}_{\geq 0})},$$

where J is the flip $(Jf)_k = f_{-k-1}$ on $\ell^2(\mathbb{Z})$. In matrix form,

$$H(a)_{ij} = a_{i+j+1} \quad (i, j \geq 0),$$

so it is constant along anti-diagonals.

A fundamental algebraic identity relating these objects is

$$(15) \quad T(a)T(b) = T(ab) + H(a)H(\tilde{b}), \quad \tilde{b}(z) := b(z^{-1}),$$

valid under mild summability assumptions (e.g. $a, b \in \mathcal{W}$). The additional Hankel term measures the failure of the Toeplitz compression to be multiplicative.

0.3. Wiener–Hopf factorization and invertibility of Toeplitz operators. Write the Fourier decomposition

$$a(z) = a_-(z) a_+(z),$$

where

$$a_+(z) = \sum_{k \geq 0} a_k^{(+)} z^k, \quad a_-(z) = \sum_{k \leq 0} a_k^{(-)} z^k,$$

and both a_+ and a_- are nonvanishing on $\mathbb{T}$. Such a factorization is called a *Wiener–Hopf factorization*. In the Wiener algebra $\mathcal{W}$ one can formulate it as

$$a \in \mathcal{W}, \quad a(z) \neq 0 \quad \forall z \in \mathbb{T}, \quad \text{wind}(a, 0) = 0 \quad \implies \quad a = a_- a_+, \quad a_{\pm}^{\pm 1} \in \mathcal{W},$$

where $\text{wind}(a, 0)$ is the winding number of $a(\mathbb{T})$ around 0. When $\text{wind}(a, 0) \neq 0$, a canonical factorization includes an extra monomial z^{κ} .

When $a = a_- a_+$ with $a_{\pm}^{\pm 1} \in \mathcal{W}$ and $\text{wind}(a, 0) = 0$, the Toeplitz operator $T(a)$ is invertible on $\ell^2(\mathbb{Z}_{\geq 0})$, and one has the operator identity

$$(16) \quad T(a)^{-1} = T(a_+^{-1}) T(a_-^{-1}) \quad \text{modulo a compact correction (in fact Hankel-type)}.$$

More precisely, using (15) one finds that $T(a_+^{-1})T(a_-^{-1})$ is a two-sided inverse of $T(a)$ up to a correction expressed through Hankel operators built from $a_{\pm}^{\pm 1}$. This is the analytic core of the Wiener–Hopf method for Toeplitz operators.

0.4. Finite Toeplitz matrices and approximate inverses as $n \rightarrow \infty$.

Even when $T(a)$ is easy to invert at the operator level, the explicit inverse of the finite matrix $T_n(a)$ can be difficult. A useful viewpoint is that $T_n(a)$ is a finite section of $T(a)$, and $T_n(a)^{-1}$ should be close to the corresponding finite section of $T(a)^{-1}$, up to boundary effects near 0 and $n-1$.

Assume $a \in \mathcal{W}$ is nonvanishing on $\mathbb{T}$ with $\text{wind}(a, 0) = 0$, and choose a Wiener–Hopf factorization $a = a_- a_+$ with $a_{\pm}^{\pm 1} \in \mathcal{W}$. Then the following principles are standard.

(1) Finite section method. Under the assumptions above, the finite sections are stable:

$$T_n(a) \text{ is invertible for all sufficiently large } n, \quad \sup_{n \geq n_0} \|T_n(a)^{-1}\| < \infty.$$

Moreover, in a strong sense,

$$T_n(a)^{-1} \approx P_n T(a)^{-1} P_n \quad (n \rightarrow \infty),$$

with an error concentrated near the boundary (hence effectively low-rank after suitable truncations).

(2) Gohberg–Semencul-type structure. There are explicit representations of $T_n(a)^{-1}$ in terms of Toeplitz and Hankel matrices built from the Fourier coefficients of $a_{\pm}^{\pm 1}$, showing that

$$T_n(a)^{-1} = T_n(a_+^{-1}) T_n(a_-^{-1}) + (\text{boundary correction}),$$

where the boundary correction has small rank compared to n (often rank at most $2r$ for symbols with finite bandwidth, and more generally a correction whose operator norm decays as $n \rightarrow \infty$ under additional regularity/analyticity).

(3) Exponentially small errors for analytic symbols. If a extends analytically and nonvanishingly to an annulus $\{r < |z| < R\}$ with $r < 1 < R$, then the Fourier coefficients of $a_{\pm}^{\pm 1}$ decay exponentially, and the boundary correction above becomes exponentially small in n (in operator norm or entrywise, depending on the formulation). This is the regime in which approximate inverses are particularly effective.

In applications, one typically combines the operator factorization (which yields a manageable expression for $T(a)^{-1}$) with one of the finite- n approximation principles above. This allows one to control kernels or partition functions in the limit $n \rightarrow \infty$ without requiring an explicit closed form for $T_n(a)^{-1}$.

0.5. The special case of rational symbols. An important and particularly tractable situation arises when the symbol $a(z)$ is a rational function of z . In this case many of the abstract constructions discussed above admit explicit realizations, and several difficulties present for general symbols disappear.

Assume that $a(z)$ is rational and has no zeros or poles on the unit circle. Then a automatically belongs to the Wiener algebra $\mathcal{W}$, and its Wiener–Hopf factorization can be carried out explicitly by separating the zeros and poles inside and outside the unit disk. More precisely, one can write

$$a(z) = c z^{\kappa} \prod_{j=1}^J \frac{1 - \overline{\alpha_j} z}{1 - \alpha_j z} \prod_{k=1}^K \frac{1 - \beta_k z^{-1}}{1 - \overline{\beta_k} z^{-1}},$$

where $|\alpha_j| < 1$, $|\beta_k| < 1$, $c \neq 0$, and $\kappa \in \mathbb{Z}$ is the winding number of a . From this representation the Wiener–Hopf factors a_+ and a_- are obtained by grouping together the factors that are analytic inside and outside the unit disk, respectively.

In the rational case the associated Toeplitz and Hankel operators have additional structure. In particular, the Hankel operators $H(a)$ and $H(\tilde{a})$ have finite rank, with rank equal to the total number of poles of a inside or outside the unit disk. As a consequence, the correction terms appearing in identities such as

$$T(a)^{-1} = T(a_+^{-1}) T(a_-^{-1}) + (\text{Hankel correction})$$

are finite-rank operators.

This finite-rank property has two important implications. First, the inverse Toeplitz operator $T(a)^{-1}$ differs from the simple product $T(a_+^{-1})T(a_-^{-1})$ only by an explicitly computable, finite-dimensional correction. Second, for finite Toeplitz matrices $T_n(a)$, the boundary corrections in the approximate inverse are uniformly finite rank, independent of n . In particular, away from the boundary the entries of $T_n(a)^{-1}$ stabilize exactly once n is sufficiently large.

In asymptotic problems, and especially in determinantal point process limits, rational symbols therefore lead to kernels that are finite-rank perturbations of translation-invariant kernels. This feature is often crucial for explicit computations and for identifying integrable structures in the limiting process.

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